

Course Syllabus

S&DS 6190: Sequential Decision Making **Theoretical Foundations and Modern Applications**

Fall 2026

Tuesdays, 9:25-11:20 AM

Class location: WTS A74

Instructor: Yuejie Chi

Email: yuejie.chi@yale.edu | Office hours: Tuesdays 11:30 AM – 12:30pm

Course Description

Sequential decision making in the face of uncertainty has garnered growing interest in recent years, with successes in a multitude of applications such as recommendation systems, robotics, and AI systems. This course aims to cover important theoretical and algorithmic foundations under the paradigms of bandits and reinforcement learning (RL) and discuss research issues arising from their modern applications in generative AI. We will cover multiple important topics including stochastic bandits, adversarial bandits, planning in Markov decision processes, online and offline RL, policy optimization, and multi-agent RL, gravitating our discussions around issues such as sample complexity, computational efficiency, and function approximation. We will also illustrate how RL is used to enable reasoning and alignment of foundation models, along with related (open) research questions.

Course Format

The class will meet once a week. Class meetings will combine lectures, discussion, in-class quizzes student presentations. A substantial final project (which runs through the semester) will give students an opportunity to pursue an original research direction related to modern reinforcement learning and sequential decision making.

Course announcements, readings, and logistics will be posted on the course website. Gradescope will be used for project submission unless otherwise noted.

Enrollment, Selection Process, and Notification

This is a graduate-level course geared towards research-oriented exploration, and therefore some level of maturity is expected. Enrollment is open to graduate students and advanced undergraduates with the necessary mathematical and coding background, subject to classroom capacity and Yale registration policies. Students who are unsure whether their preparation is appropriate should contact the instructor early in the term.

Enrollment is by permission only. Students interested in this course should submit a statement of interest, describing their background in relevant courses. This should be submitted on Canvas in the "Assignments" area. You will be informed within a week after the first meeting of the class.

Prerequisites

Students should have basic fluency in linear algebra, probability, optimization, machine learning, and programming. Prior exposure to reinforcement learning is helpful but not required.

Course Materials

There is no required textbook to purchase. Required papers and lecture notes will be posted on the course website. The following books and notes will be useful references throughout the semester:

- Alekh Agarwal, Nan Jiang, Sham M. Kakade, and Wen Sun, Reinforcement Learning: Theory and Algorithms.
- Richard S. Sutton and Andrew G. Barto, Reinforcement Learning: An Introduction.
- Tor Lattimore and Csaba Szepesvari, Bandit Algorithms.
- Yuxin Chen, Yuejie Chi, Jianqing Fan, Gen Li, Yuting Wei, and Yuling Yan, Sample Complexity of Reinforcement Learning: A Non-Asymptotic Perspective.

Grading

Your grade in the class will be a weighted average of the following components:

Component	Weight
Participation and attendance	5%
In-class quizzes	15%
Midterm presentation	10%
Project development: proposal, checkpoints, peer review	35%
Final deliverables and report	35%

The Use of LLMs

You are encouraged use LLMs to help with writing, understanding the lecture materials, brainstorming, and coding. However, you own your submitted/posted work, and take full responsibility in its consequences – so verify critically and be ready to defend your work when prompted. You are also required to disclose how LLMs are used in your submitted work.

Course Schedule

This is a provisional schedule and may be updated as the semester proceeds.

Week	Lecture	Topic	Deliverables
1	Sep 8	Introduction and logistics; bandits	
2	Sep 15	Bandits continued	
3	Sep 22	MDP and dynamic programming	Project proposal 1 st draft
4	Sep 29	Simulator setting: evaluation and planning	
5	Oct 6	Online RL	Project proposal 2 nd draft
6	Oct 13	Imitation learning and offline RL	Midterm presentation draft
7	Oct 20	Midterm presentation and peer feedback	
8	Oct 27	Policy optimization in RL	Project proposal final version
9	Nov 3	Multi-agent RL (MARL)	
10	Nov 10	RLHF	Project 1 st milestone
11	Nov 17	RLVR	
Break	Nov 24	No class: November recess	Project 2 nd milestone
12	Dec 1	Guest lecture on RL practice (tentative)	Final presentation draft due
13	Dec 8	Final project presentation	
Finals	Dec 15-22	Final project report due	

Questions?

Please write to yuejie.chi@yale.edu for questions!